

Numerical Differentiation

Advance Math

Informatika, FTI UII

July 7, 2022

Recall that the derivative of the function f at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad (1)$$

which gives an obvious way to approximate $f'(x)$ by computing

$$\frac{f(x+h) - f(x)}{h},$$

for small values of h .

For several reasons, we need to **approximate** derivatives, instead, and thus

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}. \quad (2)$$

Using the Taylor expansion of $f(x+h)$ about x , we write

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(\xi), \quad \xi \in (x, x+h), \quad (3)$$

assuming that f has two continuous derivatives.

With the Taylor expansion in (3), we can replace the approximation in (2) with an exact formula that reads

$$f'(x) = \frac{f(x+h) - f(x)}{h} - \underbrace{\frac{h}{2}f''(\xi)}_{\text{error term}}. \quad (4)$$

This formula is called the **forward difference** if $h > 0$ and the **backward difference** if $h < 0$.

Hence, it is reasonable to use the approximation

$$f'(x) \approx \frac{f(x+h) - f(x)}{h},$$

with the *truncation* error involved is

$$|e| = \frac{h}{2} |f''(\xi)| \leq \max_{t \in (x, x+h)} |f''(t)|. \quad (5)$$

Exercise

Use forward difference to approximate the derivative of $f(x) = \ln x$ at $x_0 = 1.8$ with $h = 0.1, h = 0.05, h = 0.01$. Determine the bound of the errors.

Exercise

Write in terms of the forward difference formula, we get

$$\frac{f(1.8 + h) - f(1.8)}{h}.$$

With $h = 0.1$,

$$\frac{\ln 1.9 - \ln 1.8}{0.1} = \frac{0.64185389 - 0.58778667}{0.1} = 0.5406722.$$

As $f''(x) = -1/x^2$ and $1.8 < \xi < 1.9$, a bound for this approximation is

$$|e| = \frac{h}{2} |f''(\xi)| = \frac{|h|}{2\xi^2} \leq \frac{0.1}{2(1.8)^2} = 0.0154321.$$

Continue yourself when $h = 0.05$ and $h = 0.01$.

Solution

h	$f(1.8 + h)$	$\frac{f(1.8 + h) - f(1.8)}{h}$	$\frac{ h }{2(1.8)^2}$
0.1	0.64185389	0.5406722	0.0154321
0.05	0.61518564	0.5479795	0.0077160
0.01	0.59332685	0.5540180	0.0015432

Since $f(x) = 1/x$, the exact value of $f(1.8)$ is $0.5\overline{55}$, and in this case the error bounds are quite close to the true approximation error.

Three-point formulas

Let x_0, x_1 and x_2 be **equally spaced**, that is,

$$x_1 = x_0 + h, \quad x_2 = x_0 + 2h, \quad \text{for some } h \neq 0.$$

The **three-point endpoint** formula reads

$$f'(x_0) = \frac{1}{2h}[-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \frac{h^2}{3}f^{(3)}(\xi_0). \quad (6)$$

where ξ_0 lies between x_0 and $x_0 + 2h$.

The **three-point midpoint** formula reads

$$f'(x_0) = \frac{1}{2h}[f(x_0 + h) - f(x_0 - h)] + \frac{h^2}{3}f^{(3)}(\xi_1). \quad (7)$$

where ξ_1 lies between $x_0 - h$ and $x_0 + h$.

Exercise

Values for $f(x) = xe^x$ are given by the table below

x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030

Use all the applicable three-point formulas in (6) and (7) to approximate $f'(2.0)$.

Use the endpoint formula (6), with $x_0 = 2.0$, $h = 0.1$, i.e.,

$$\begin{aligned} f'(x_0) &= \frac{1}{2h}[-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \underbrace{\frac{h^2}{3}f^{(3)}(\xi_0)}_{\text{error term}} \\ &= \frac{1}{0.2}[-3f(2.0) + 4f(2.1) - f(2.2)] \\ &= 5[-3(14.778112) + 4(17.148957) - 19.855030)] \\ &= 22.032310. \end{aligned}$$

Typically we compute the error term separately, thus we ignore it at this moment.

Exercise

Show the result of

- 1 endpoint formula with $h = -0.1$
- 2 midpoint formula with $h = 0.1$
- 3 midpoint formula with $h = 0.2$

"Numerical analysis" by Richard L. Burden and J. Douglas Faires

"Numerical analysis" by L. Ridgway Scott